

Class activity 3:

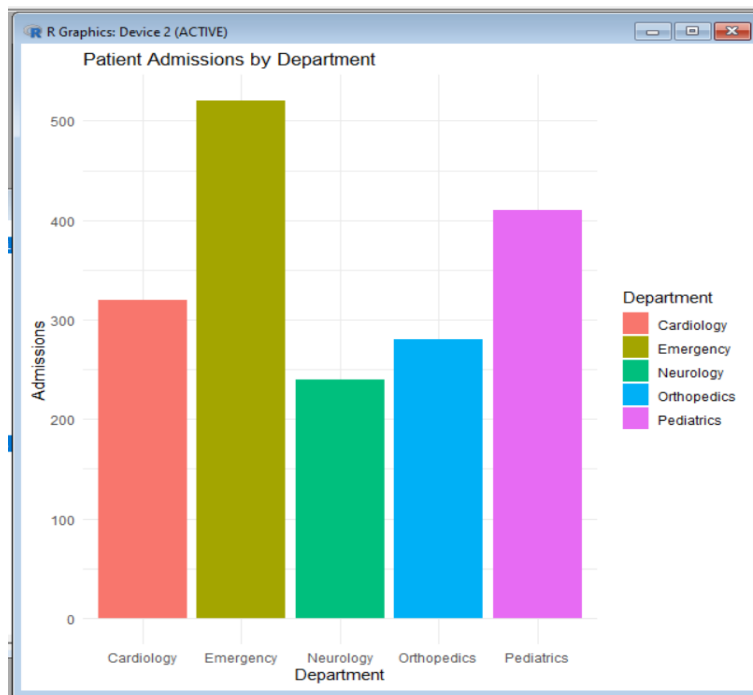
1. Hospital Patient Admission Analysis:

Bar chart:

Python code:

```
hospital <- data.frame(  
  Department = c("Cardiology", "Neurology", "Pediatrics", "Orthopedics", "Emergency"),  
  Admissions = c(320, 240, 410, 280, 520),  
  Male = c(170, 120, 210, 150, 280),  
  Female = c(150, 120, 200, 130, 240),  
  StayDays = c(5, 6, 3, 4, 2)  
)  
ggplot(hospital,  
  aes(x = Department,  
    y = Admissions,  
    fill = Department)) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Patient Admissions by Department",  
    x = "Department",  
    y = "Admissions"  
  ) +  
  theme_minimal()
```

Output:



Stacked Bar chart:

Python code:

```
library(tidyr)
```

```
hospital_long <- pivot_longer(  
  hospital,  
  cols = c(Male, Female),  
  names_to = "Gender",  
  values_to = "Count"  
)
```

```
ggplot(hospital_long,  
  aes(x = Department,  
    y = Count,  
    fill = Gender)) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Admissions by Gender",  
    x = "Department",  
    y = "Patients"  
  )
```

Output:

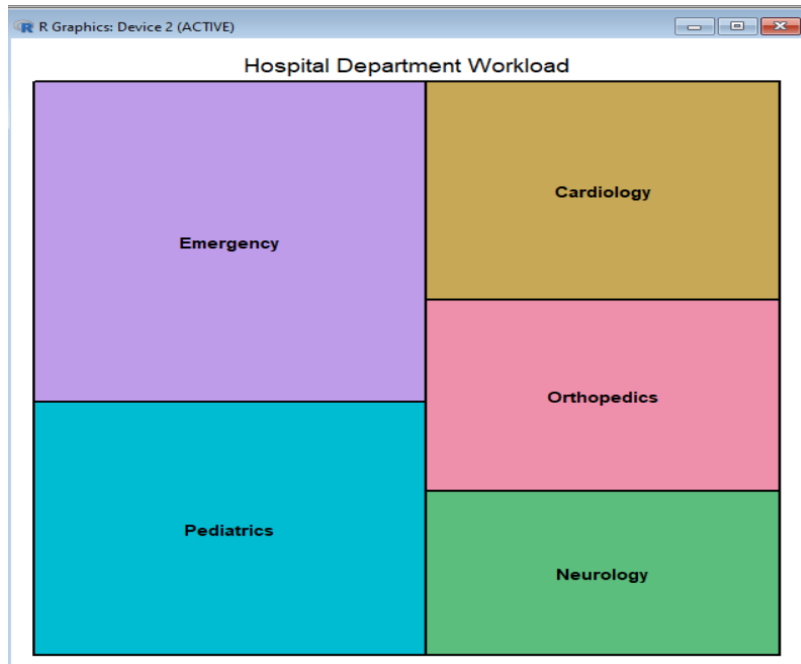


Tree map

Python code:

```
treemap(  
    hospital,  
    index = "Department",  
    vSize = "Admissions",  
    title = "Hospital Department Workload"  
)
```

Output:

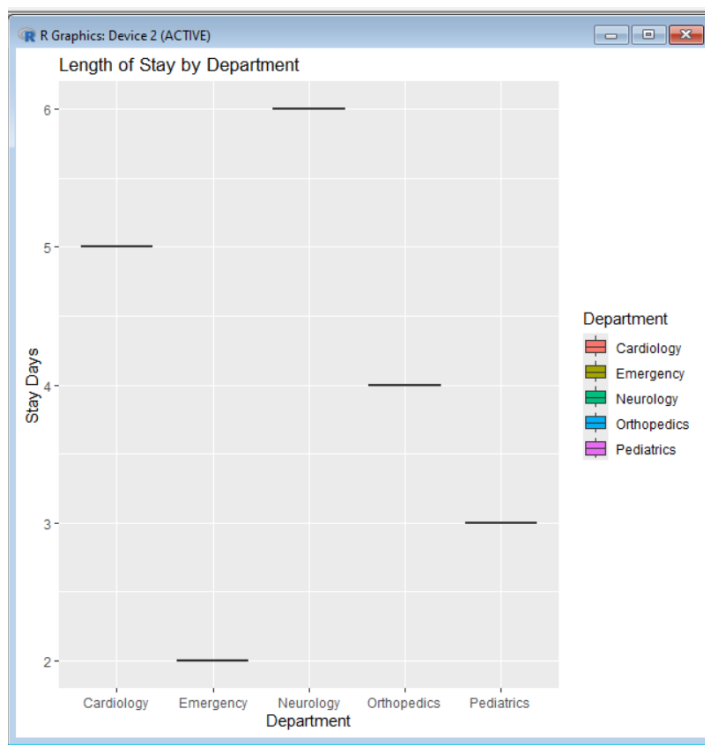


Box plot:

Python code:

```
ggplot(hospital,  
    aes(x = Department,  
        y = StayDays,  
        fill = Department)) +  
geom_boxplot() +  
labs(  
    title = "Length of Stay by Department",  
    x = "Department",  
    y = "Stay Days"  
)
```

Output:

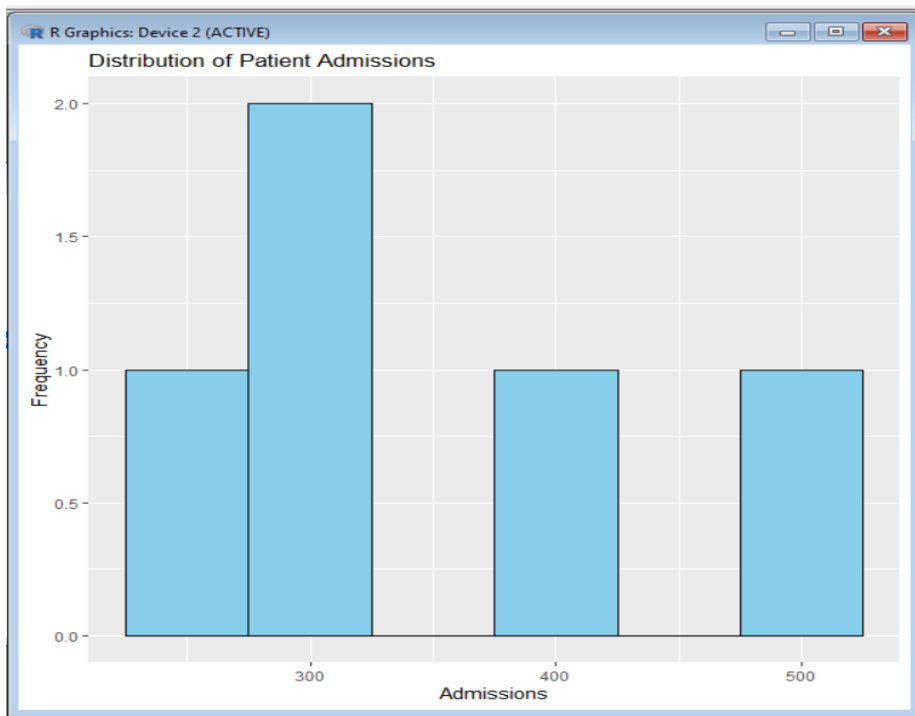


Histogram

Python code:

```
ggplot(hospital,
       aes(x = Admissions)) +
  geom_histogram(
    binwidth = 50,
    fill = "skyblue",
    color = "black"
  ) +
  labs(
    title = "Distribution of Patient Admissions",
    x = "Admissions",
    y = "Frequency"
  )
```

Output:



2. E-Commerce Customer Spending Behaviour

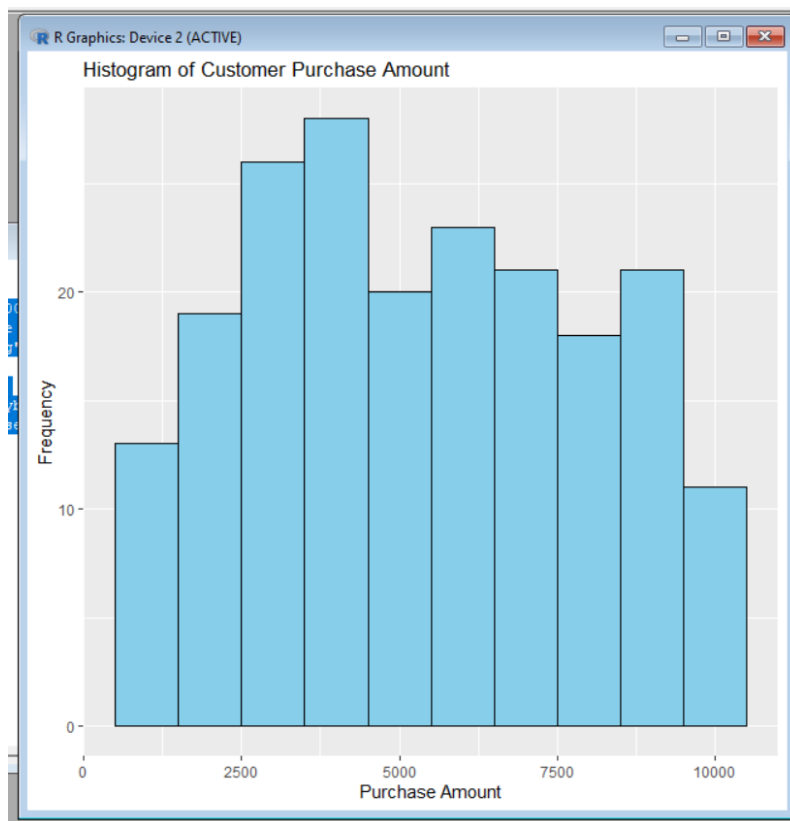
Histogram:

Python code:

```
set.seed(123)
customer <- data.frame(
  Purchase_Amount = round(runif(200, 200, 10000), 0),
  Order_Frequency = sample(1:20, 200, replace = TRUE),
  Category = sample(c("Electronics", "Clothing", "Grocery", "Home", "Sports"), 200,
    replace = TRUE)
)
ggplot(customer, aes(x = Purchase_Amount)) +
  geom_histogram(binwidth = 1000, fill = "skyblue", color = "black") +
  labs(title = "Histogram of Customer Purchase Amount",
    x = "Purchase Amount",
    y = "Frequency")

y = "Number of Customers")
```

Output:



Violin plot:

Python code:

```
ggplot(customer, aes(x = Category, y = Purchase_Amount, fill = Category)) +  
geom_violin() + labs(title = "Violin Plot of Purchase Amount by Category", x =  
"Category", y = "Purchase Amount")
```

Output:



Box plot:

Python code:

```
ggplot(customer, aes(x = Category, y = Purchase_Amount, fill = Category)) +  
geom_boxplot() + labs(title = "Box Plot of Purchase Amount by Category", x =  
"Category", y = "Purchase Amount")
```

Output:

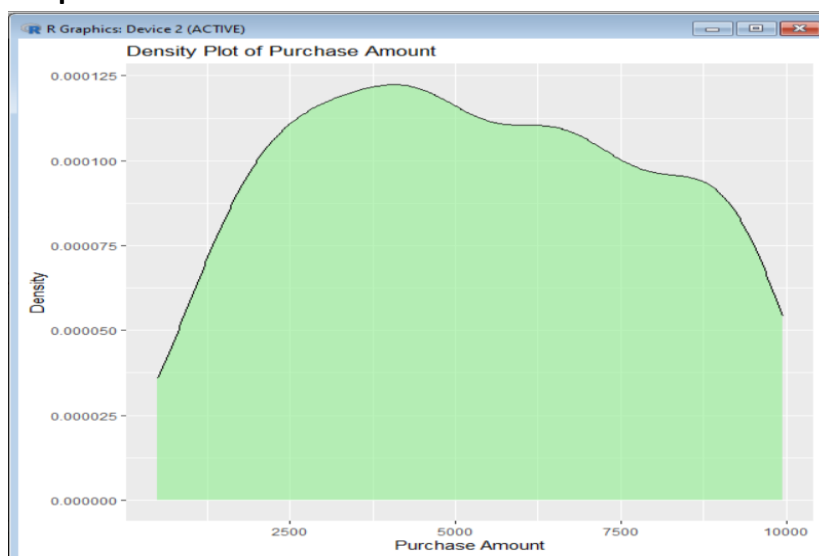


Density plot

Python code:

```
ggplot(customer, aes(x = Purchase_Amount)) + geom_density(fill = "lightgreen",  
alpha = 0.6) + labs(title = "Density Plot of Purchase Amount", x = "Purchase Amount",  
y = "Density")
```

Output:



Bar plot

Python code:

```
ggplot(customer, aes(x = Category, fill = Category)) + geom_bar() + labs(title = "Customer Count by Product Category", x = "Product Category", y = "Number of Customers")
```

Output:

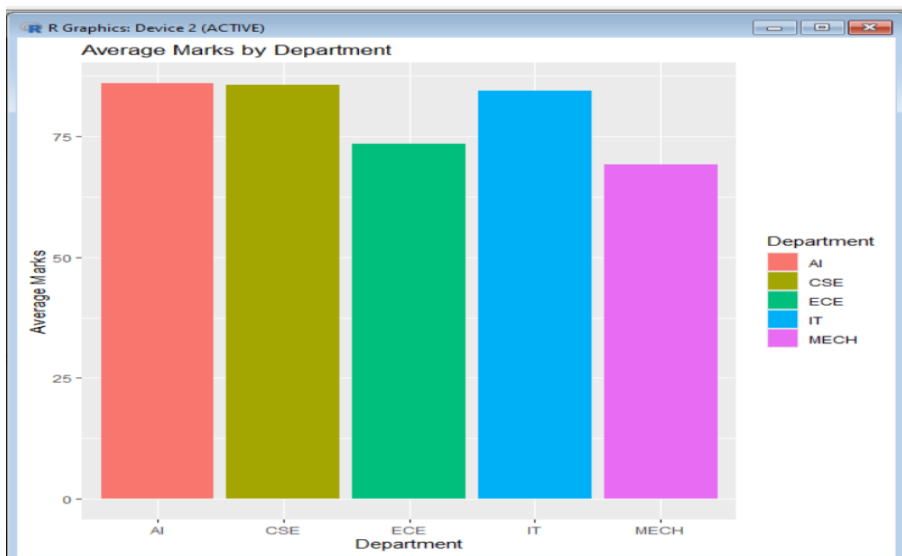


3. University Examination Performance Dashboard

Bar plot:

```
ggplot(exam, aes(x=Department, y=Marks, fill=Department)) +  
geom_bar(stat="summary", fun="mean") + labs(title="Average Marks by  
Department", x="Department", y="Average Marks")
```

Output:

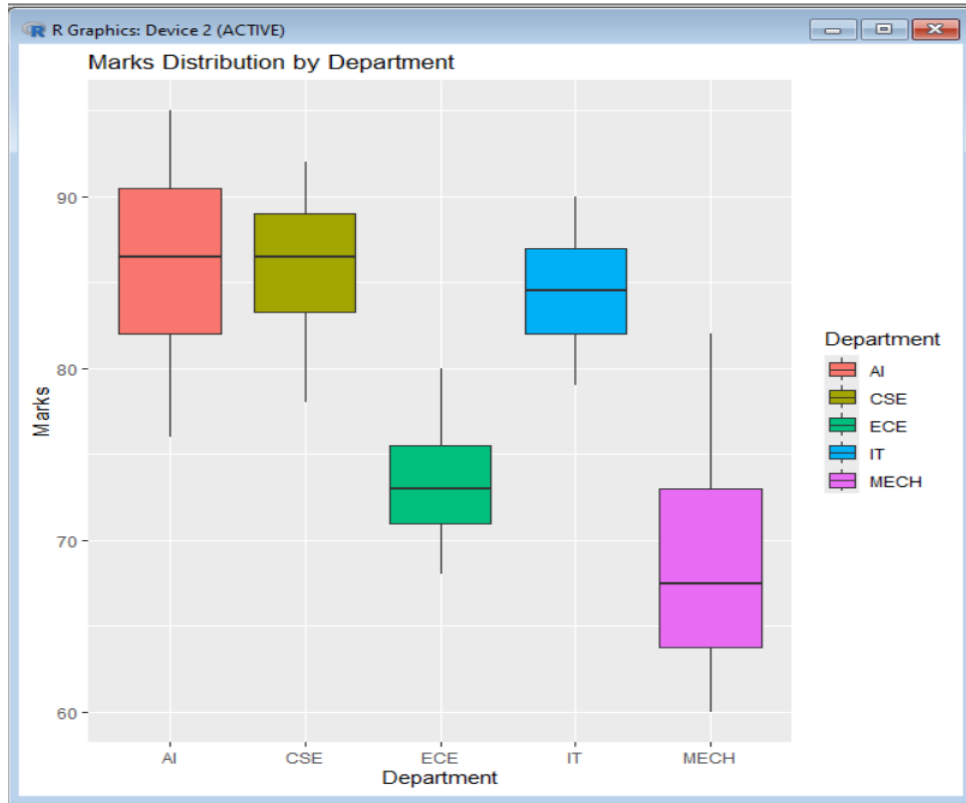


Box plot

Python code:

```
ggplot(exam, aes(x=Department, y=Marks, fill=Department)) + geom_boxplot() +  
labs(title="Marks Distribution by Department", x="Department", y="Marks")
```

Output:

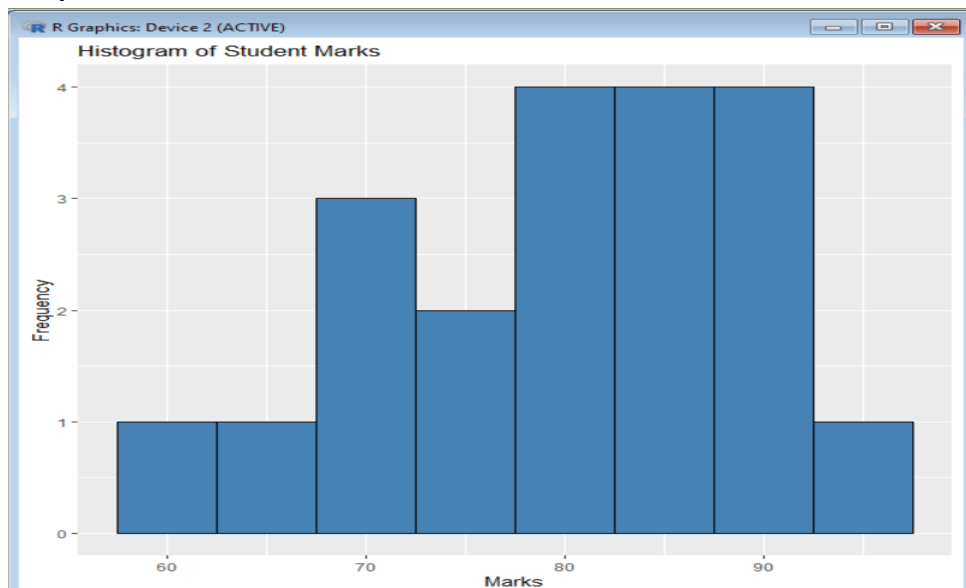


Histogram

Python code:

```
ggplot(exam, aes(x=Marks)) + geom_histogram(binwidth=5, fill="steelblue",  
color="black") + labs(title="Histogram of Student Marks", x="Marks", y="Frequency")
```

Output:

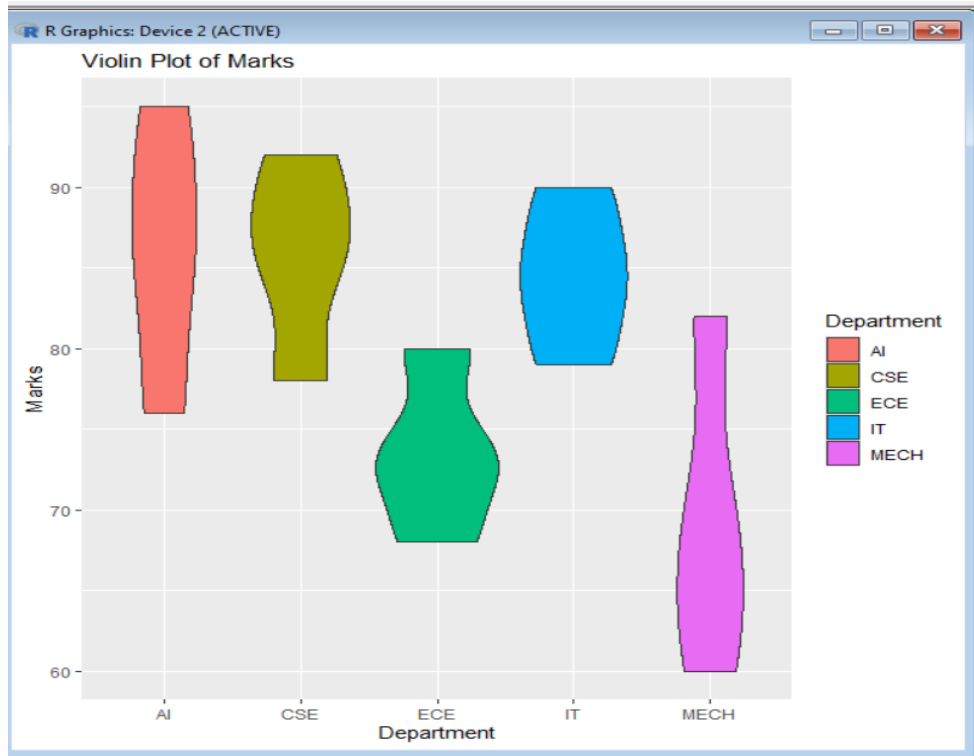


Violin plot

Python code:

```
ggplot(exam, aes(x=Department, y=Marks, fill=Department)) + geom_violin() +  
labs(title="Violin Plot of Marks", x="Department", y="Marks")
```

Output:

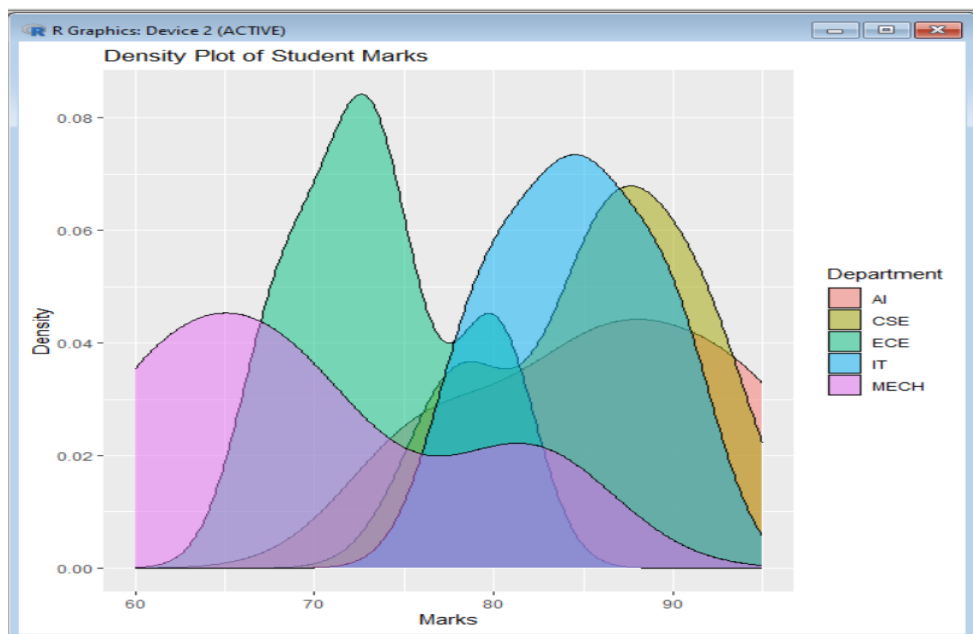


Density plot

Python code:

```
ggplot(exam, aes(x=Marks, fill=Department)) + geom_density(alpha=0.5) +  
labs(title="Density Plot of Student Marks", x="Marks", y="Density")
```

Output:



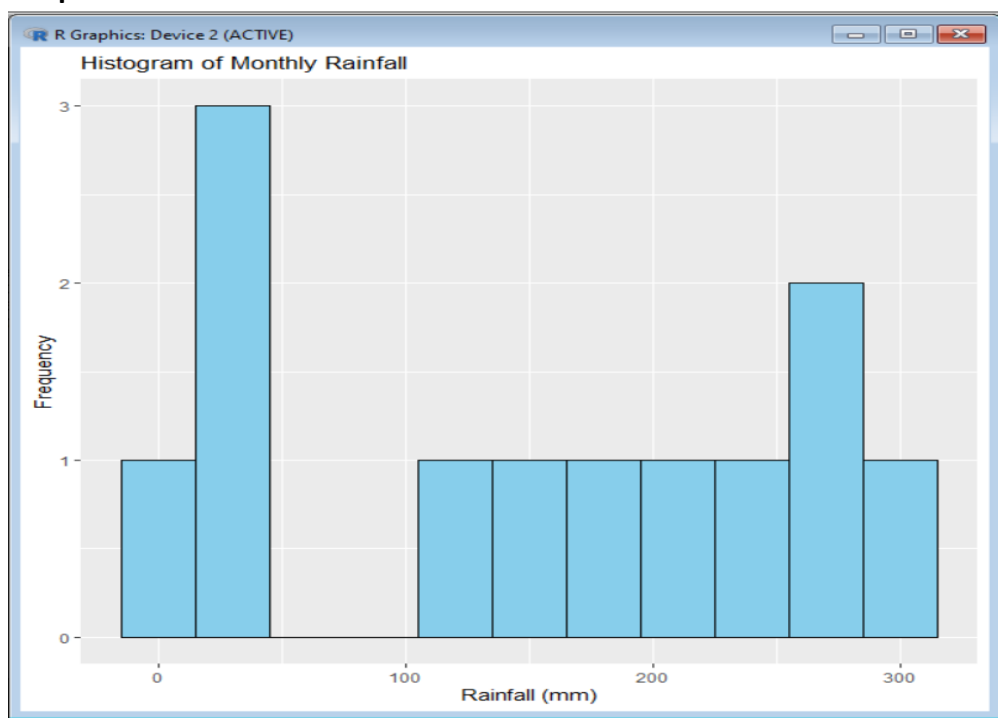
4. Climate and Rainfall Distribution Study

Histogram

Python code:

```
climate <- data.frame(  
  Month = month.abb,  
  City = rep(c("Delhi", "Mumbai", "Chennai"), each = 4),  
  Rainfall = c(12, 18, 30, 45, 180, 250, 310, 280, 120, 160, 210, 260),  
  Temperature = c(18, 22, 28, 35, 30, 29, 28, 27, 32, 34, 33, 31),  
  Humidity = c(45, 50, 55, 60, 75, 82, 88, 85, 70, 72, 78, 80)  
)  
  
ggplot(climate, aes(x=Rainfall)) +  
  geom_histogram(binwidth=30,  
  fill="skyblue",  
  color="black") +  
  labs(title="Histogram of Monthly Rainfall",  
  x="Rainfall (mm)",  
  y="Frequency")
```

Output:

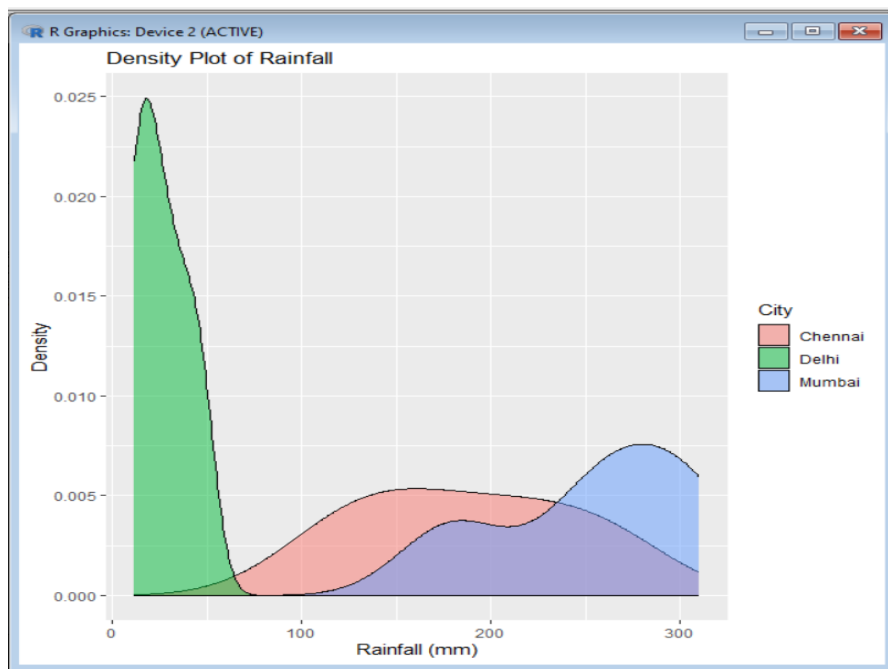


Density plot

Python code:

```
ggplot(climate, aes(x=Rainfall, fill=City)) + geom_density(alpha=0.5) +  
  labs(title="Density Plot of Rainfall", x="Rainfall (mm)", y="Density")
```

Output:

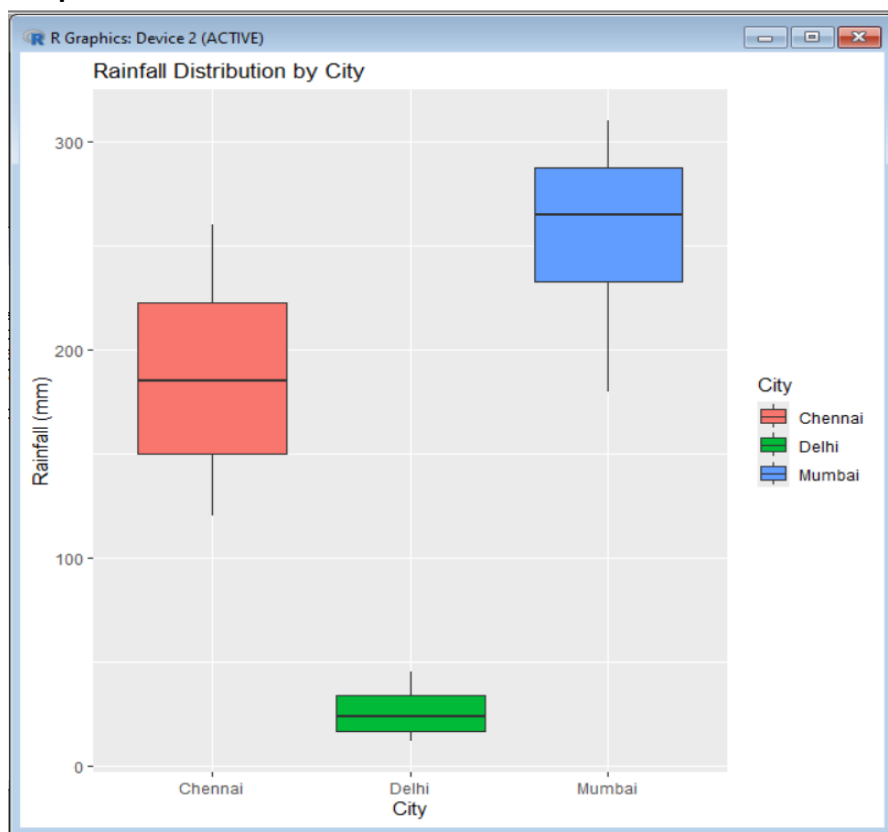


Box plot

Python code:

```
ggplot(climate, aes(x=City, y=Rainfall, fill=City)) + geom_boxplot() +  
labs(title="Rainfall Distribution by City", x="City", y="Rainfall (mm)")
```

Output:

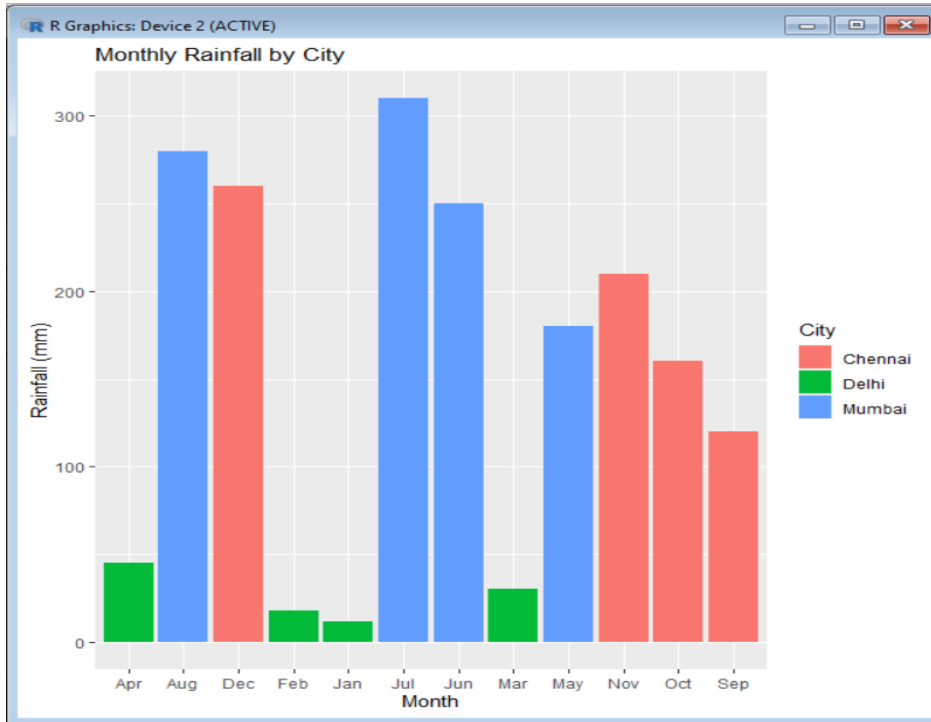


Bar plot

Python code:

```
ggplot(climate, aes(x=Month, y=Rainfall, fill=City)) + geom_bar(stat="identity",  
position="dodge") + labs(title="Monthly Rainfall by City", x="Month", y="Rainfall  
(mm)")
```

Output:



5. Bank Loan Risk Assessment

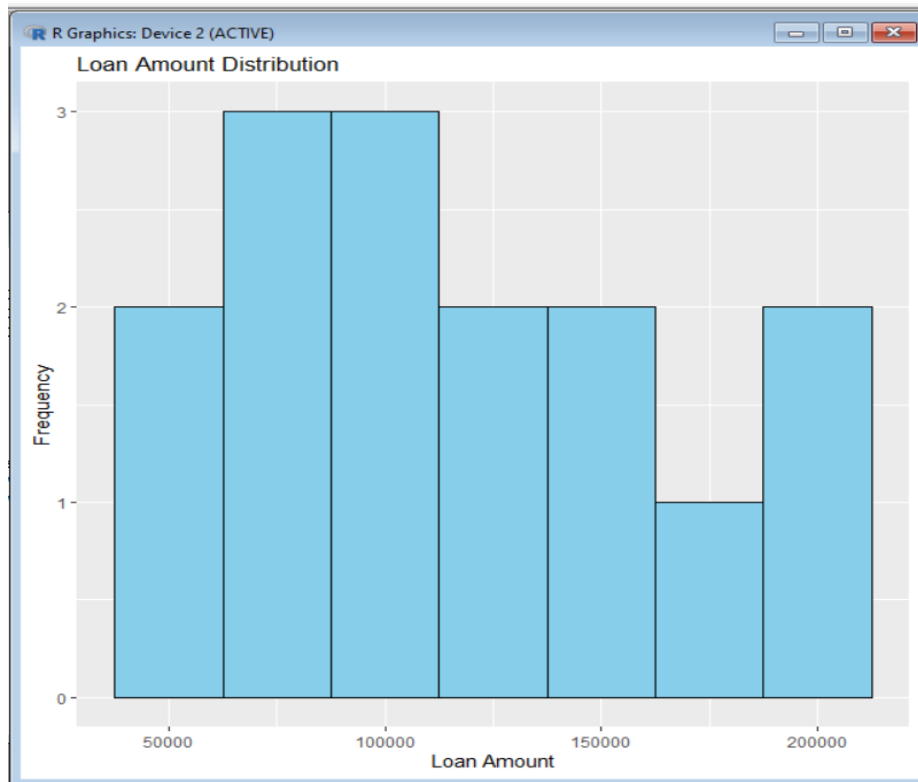
Histogram

Python code:

```
loan <- data.frame(  
  Customer = paste("C", 1:15, sep=""),  
  LoanAmount = c(50000,75000,120000,90000,150000,  
    200000,80000,60000,170000,110000,  
    95000,130000,210000,70000,160000),  
  Income = c(30000,45000,55000,40000,70000,  
    85000,50000,35000,90000,60000,  
    48000,65000,95000,42000,80000),  
  CreditScore = c(620,680,720,650,780,  
    810,690,640,790,710,  
    670,740,820,660,760),  
  Repayment = c("Poor","Average","Good","Average","Good",  
    "Good","Average","Poor","Good","Good",  
    "Average","Good","Good","Poor","Good")  
)  
ggplot(loan, aes(x=LoanAmount)) +
```

```
geom_histogram(binwidth=25000,
fill="skyblue",
color="black") +
labs(title="Loan Amount Distribution",
x="Loan Amount",
y="Frequency")
```

Output:

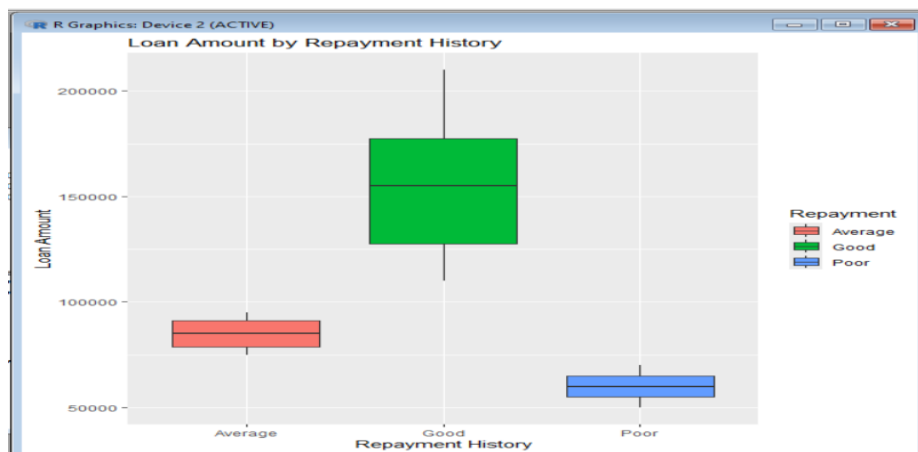


Box plot

Python code:

```
ggplot(loan, aes(x=Repayment, y=LoanAmount, fill=Repayment)) + geom_boxplot() +
labs(title="Loan Amount by Repayment History", x="Repayment History", y="Loan Amount")
```

Output:



Scatter plot

Python code:

```
ggplot(loan, aes(x=Income, y=LoanAmount, color=CreditScore)) +  
geom_point(size=4) + scale_color_gradient(low="red", high="green") +  
labs(title="Income vs Loan Amount", x="Income", y="Loan Amount")
```

Output:



Scatter plot

Python code:

```
ggplot(loan, aes(x=Repayment, fill=Repayment)) + geom_bar() +  
labs(title="Customers by Repayment History", x="Repayment History", y="Number of  
Customers")
```

Output:

